

Processando config 23, seed 2, teste: BlockAlphabit
xx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xx

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 0.052
p-value of test : 0.97

CPU time used : 00:00:00.03

Generator state:
16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 9.16
p-value of test : 0.33

CPU time used : 00:00:00.02

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 134.26
p-value of test : 0.33

CPU time used : 00:00:00.02

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33035.09
p-value of test : 0.15

CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 142.09
p-value of test : 0.97

CPU time used : 00:00:00.03

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 305.57
p-value of test : 0.48

CPU time used : 00:00:00.02

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.57
p-value of test : 0.71

CPU time used : 00:00:00.02

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.24
p-value of test	:	0.75

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	32.77
p-value of test	:	0.38

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	34.78
p-value of test	:	0.34

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	21.43
p-value of test	:	0.67

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	13.50
p-value of test	:	0.56

CPU time used : 00:00:00.08

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	57.26
p-value of test	:	0.65

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	74.65
p-value of test	:	0.13

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	162.63
p-value of test	:	0.43

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	54.23
p-value of test	:	0.58

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	29.27
p-value of test	:	0.61

CPU time used : 00:00:00.06

Generator state:
150994944 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.31

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 2.58
p-value of test                   : 0.28
```

```
-----
CPU time used                     : 00:00:00.02
```

```
Generator state:
```

```
167772160 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
```

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	8
Value of the statistic	:	5.49
p-value of test	:	0.70

CPU time used : 00:00:00.02

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	128
Value of the statistic	:	127.58
p-value of test	:	0.49

CPU time used : 00:00:00.03

Generator state:

201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 0.00097653271, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 32768
Value of the statistic : 32535.85
p-value of test : 0.82
```

```
-----
CPU time used : 00:00:00.03
```

Generator state:

218103808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

sstring_HammingIndep test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom : 177
Chi-square statistic : 176.62
p-value of test : 0.49
```

```
-----
CPU time used : 00:00:00.02
```

Generator state:

234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	317.53
p-value of test	:	0.30

CPU time used	:	00:00:00.00
---------------	---	-------------

Generator state:

251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	2.18
p-value of test	:	0.01

CPU time used	:	00:00:00.03
---------------	---	-------------

Generator state:

268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	40.07
p-value of test	:	0.15

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	25.42
p-value of test	:	0.75

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	35.50
p-value of test	:	0.31

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	17.82
p-value of test	:	0.85

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	9.41
p-value of test	:	0.86

CPU time used : 00:00:00.06

Generator state:

285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	56.86
p-value of test	:	0.66

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	61.43
p-value of test	:	0.50

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	167.80
p-value of test	:	0.32

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	52.55
p-value of test	:	0.64

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	54.55
p-value of test	:	7.7e-3

CPU time used : 00:00:00.05

Generator state:

301989888 bits have been read.

===== Summary results of Alhabit =====

Version:	TestU01 1.2.3
Generator:	ufire_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.28

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 5.36
p-value of test                   : 0.07
```

```
-----
CPU time used                      : 00:00:00.02
```

```
Generator state:
318767104 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE

Number of bits = n = 16777216
```

Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 2.0861626e-07$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom : 8
Value of the statistic : 22.03
p-value of test : $4.9e-3$

CPU time used : 00:00:00.02

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: $r = 0$, $s = 32$, $w = 4$

smultin_MultinomialBitsOver test:

 $N = 1$, $n = 16777216$, $r = 0$, $s = 32$, $L = 8$,
Sparse = FALSE

Number of bits = $n = 16777216$
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

$\Delta = 1$, $\mu = 3.784895e-06$, $\sigma = 1$

Test Results for $\Delta = 1.0000$

Number of degrees of freedom : 128
Value of the statistic : 111.10
p-value of test : 0.86

CPU time used : 00:00:00.02

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32801.77
p-value of test : 0.45

CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 184.17
p-value of test : 0.34

CPU time used : 00:00:00.02

Generator state:

385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	312.66
p-value of test	:	0.37

CPU time used : 00:00:00.02

Generator state:

402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	1.76
p-value of test	:	0.04

CPU time used : 00:00:00.00

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	55.76
p-value of test	:	5.8e-3

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	62.27
p-value of test	:	7.2e-4 *****

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	36.00
p-value of test	:	0.29

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	13.81
p-value of test	:	0.96

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	16.30
p-value of test	:	0.36

CPU time used : 00:00:00.06

Generator state:

436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	58.55
p-value of test	:	0.60

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	59.66
p-value of test	:	0.56

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	144.37
p-value of test	:	0.81

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	55.21
p-value of test	:	0.54

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.37
p-value of test	:	0.75

CPU time used : 00:00:00.06

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216

Number of statistics: 17
Total CPU time: 00:00:00.27
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
8 RandomWalk1 M (L = 64)	7.2e-4

All other tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 7.93
p-value of test : 0.02

CPU time used : 00:00:00.03

Generator state:
469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 4.36
p-value of test : 0.82

CPU time used : 00:00:00.00

Generator state:

486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 117.61
p-value of test : 0.73

CPU time used : 00:00:00.02

Generator state:
503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32798.79
p-value of test : 0.45

CPU time used : 00:00:00.03

Generator state:
520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 203.46
p-value of test : 0.08

CPU time used : 00:00:00.02

Generator state:
536870656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 343.49
p-value of test : 0.06

CPU time used : 00:00:00.00

Generator state:
553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.39
p-value of test : 0.92

CPU time used : 00:00:00.00

Generator state:

570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	33.84
p-value of test	:	0.38

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	29.02
p-value of test	:	0.57

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	46.40
p-value of test	:	0.05

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	31.50
p-value of test	:	0.17

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	15.70
p-value of test	:	0.40

CPU time used : 00:00:00.06

Generator state:
587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	60.36
p-value of test	:	0.54

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	75.13
p-value of test	:	0.12

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	174.23
p-value of test	:	0.21

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	46.11
p-value of test	:	0.85

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	46.17
p-value of test	:	0.05

CPU time used : 00:00:00.05

Generator state:

603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed

XX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	3.04
p-value of test	:	0.22

CPU time used : 00:00:00.02

Generator state:
620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic : 9.19
p-value of test : 0.33
```

CPU time used : 00:00:00.02

Generator state:

637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic : 133.56
p-value of test : 0.35
```

CPU time used : 00:00:00.00

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32605.82
p-value of test : 0.74

CPU time used : 00:00:00.03

Generator state:
671088128 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----
Number of degrees of freedom      : 177
Chi-square statistic              : 195.85
p-value of test                   : 0.16
-----
```

```
-----
CPU time used                     : 00:00:00.02
-----
```

Generator state:
687865344 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

```
-----
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0
-----
```

Counters with expected numbers ≥ 10

```
-----
Number of degrees of freedom      : 305
Chi-square statistic              : 326.57
p-value of test                   : 0.19
-----
```

```
-----
CPU time used                     : 00:00:00.00
-----
```

Generator state:
704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 32
-----
```

```
-----
Normal statistic                  : 0.011
p-value of test                   : 0.50
-----
```

```
-----
CPU time used                     : 00:00:00.00
-----
```

Generator state:
721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	34.87
p-value of test	:	0.33

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	41.17
p-value of test	:	0.10

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	27.95
p-value of test	:	0.67

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	25.73
p-value of test	:	0.42

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	10.52
p-value of test	:	0.79

CPU time used : 00:00:00.06

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	52.01
p-value of test	:	0.81

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	57.65
p-value of test	:	0.63

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	173.95
p-value of test	:	0.21

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	66.71
p-value of test	:	0.18

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	30.11
p-value of test	:	0.56

CPU time used : 00:00:00.05

Generator state:

754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 1.01
p-value of test : 0.60

CPU time used : 00:00:00.02

Generator state:
771751168 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 6.43
p-value of test : 0.60

CPU time used : 00:00:00.03

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128

Value of the statistic : 99.78
p-value of test : 0.97

CPU time used : 00:00:00.00

Generator state:
805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32585.61
p-value of test : 0.76

CPU time used : 00:00:00.03

Generator state:
822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 175.85  
p-value of test                   : 0.51  
-----
```

```
-----  
CPU time used                     : 00:00:00.02  
-----
```

Generator state:

838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 283.17  
p-value of test                   : 0.81  
-----
```

```
-----  
CPU time used                     : 00:00:00.00  
-----
```

Generator state:

855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                   : 0.40  
p-value of test                   : 0.35  
-----
```

CPU time used : 00:00:00.00

Generator state:
872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	35.63
p-value of test	:	0.30

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	28.98
p-value of test	:	0.57

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	27.34
p-value of test	:	0.70

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	25.26
p-value of test	:	0.45

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	14.23
p-value of test	:	0.51

CPU time used : 00:00:00.06

Generator state:

889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	75.22
p-value of test	:	0.12

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	71.15
p-value of test	:	0.20

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	218.58
p-value of test	:	1.4e-3

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	61.37
p-value of test	:	0.32

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	23.24
p-value of test	:	0.87

CPU time used : 00:00:00.05

Generator state:
905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed